

Research papers

An integrated state-estimation framework for interdependent water and energy systems

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ABSTRACT

Recent studies have shown indisputable evidence indicating water and energy networks are two interdependent cyber-physical networks and should be operated cooperatively to maximize their efficiency and resiliency. However, current state-estimation models of water and energy infrastructures are implemented independently in their respective control centers. Such independent state-estimation in energy or water network might not be able to detect false data injections in the other network. In this paper, an integrated state-estimation framework is proposed to process the consistency of the data and estimate the parameters of a combined water and energy network. Compared to standalone state-estimation, an integrated framework proposed in this work makes it more difficult for the attackers to launch stealthy false data injections, as more parameters are involved in the state-estimation that need to be modified for a successful attack. The findings of this research confirm that by accounting for interlinks between hydrological and energy systems, the security of both systems will be improved. The interconnection between the state variables and measurements of the water-energy network is reflected in the energy consuming components of the water system, and is achieved through the measurements of pump's active power consumption in the proposed integrated estimation framework. The framework includes Newton's iterative method for an alternating current (AC) state-estimation procedure for the energy network. The state-estimation of the energy network includes the estimation of voltages using measured parameters of the energy network such as active/reactive power flow/injection or voltages. Moreover, a three-tier state-estimation process based on Newton's iterative method for the water network is contemplated to attain nodal heads, nodal demands, and the flow rate of the pipes without pumps. An integrated bad data detection algorithm is also supplemented to the proposed framework to detect potential bad data in water distribution system or energy network and enhance the resilience of the integrated water energy systems.

1. Introduction

It is a well-established fact that water and energy sources are each indispensable to our modern economic systems. The water and energy sectors are increasingly under rapid reform to meet the three essentials of security of supply, sustainability, and economic efficiency. In conditions where water and energy systems rely on each other, the most challenging problems are before engineers and policymakers (Endo et al., 2017; Smajgl et al., 2016; Zeng et al., 2020). However, it will be demonstrated in this paper that such interdependence can be used towards the advantage of the two systems, to improve the accuracy of the state-estimation and bad data detection algorithms.

Cybersecurity has become one of the key challenges of the water and energy supply infrastructures in the recent years. Security agents have warned both water and energy sectors of potential false data injection (FDI), also known as cyber-physical attacks. For instance, in March and April of 2018, the Department of Homeland Security (DHS) and Federal Bureau of Investigation (FBI) apprised the water utilities of Russian government's efforts to manipulate critical infrastructure instruments involved in the water utility operations (Koehler, 2018). In 2015, a power grid in Ukraine was taken down by an attacker, and a year later, Russian hackers targeted a transmission level substation resulting in a black out at some parts of Kiev (Knake, 2017). State-estimation is a technique that is used to identify potential measurement

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errors or FDIs in water or energy networks. Using the measured data and information on the structure of the system, the parameters of energy or water networks can be estimated.

In the water networks, the measurements often include the flow of the pumps and main pipes, pressure measurements collected via telemetry systems, the total water usage recorded by district meters, and some less accurate predictions of water usage at random nodes, also known as “pseudo-measurements” (Arsene and Gabrys, 2014). These measurements are collected by smart meters installed only on the pipes, nodes, and pumps of importance, since a typical size water distribution system includes hundreds to thousands of pipes and nodes and it is economically infeasible to measure the pressure head, demand, and flow rate of every single node and pipe in the system. Accordingly, state-estimation algorithms are developed to estimate the parameters that are not recorded by smart meters yet essential to the decision making in water distribution systems. As in water systems, state-estimation is a key tool in power systems, which incorporates real and reactive power flows, bus power injections, and voltages phasor measurements to estimate voltage magnitudes and phase angles as state variables (Primadianto and Lu, 2016). Due to the high importance of state-estimation in both water and energy networks, and the challenges they introduce to both systems, various state-estimation methods in water and energy systems are discussed in the following subsection.

1.1. Literature review

Several studies have proposed state-estimation models for water distribution networks or electrical power systems (EPS) individually. Prior efforts in state-estimation in water distribution system (WDS) can be summarized into three main techniques: least square, least absolute value, and min–max methods, each of which provides some advantages with particular errors. For instance, the least-square method is known to be sensitive to large errors, whereas the least absolute value and min–max methods can cause computational issues due to their non-differentiable objective functions (Tshehla et al., 2017). For instance, the weighted least square method was used as an implicit state-estimation method for water systems with very low measurement redundancy (Andersen and Powell, 2000). A study predicted nodal demands by solving an objective function with constrained weighted-least-squares of the measured data (Tabesh et al., 2011). Others used the squared summation of residuals between the measured and estimated data as the objective function (Rozos et al., 2010). Some studies used Kalman filtering technique to enhance the water quality monitoring system (Krutys et al., 2019) or simulations (Arsene and Gabrys, 2014) to solve for the unknown parameters. In another study, the Davidon-Fletcher-Powell algorithm was used to estimate nodal demand in a WDS (Zhang et al., 2018). One of the difficulties reported in developing the state-estimation method for water systems was computational issues in pipes with small flow rates, in which the Jacobian matrix became singular (Bargiela, 1984; Arsene and Gabrys, 2014). To tackle this problem, other methods such as simulations were combined with iterative methods (Arsene and Gabrys, 2014).

Unlike in WDS where state-estimation is considered to be a modern problem, in EPS, these studies date back to over forty years ago. One of the pioneers, Schweppe (Schweppe, 1970) used a nonlinear weighted least squares optimization method in their estimation. Other studies incorporated non-iterative approaches that considered the data from Supervisory Control and Data Acquisition (SCADA) systems (Jiang et al., 2015; Dobakhshari et al., 2019). Other studies incorporated phasor measurement unit (PMU) and remote terminal unit (RTU) models into the state-estimations (Lavorato et al., 2007; Pandey et al., 2018; Jovicic et al., 2018). A recent study developed a linear model for the power system state-estimation with PMU and RTU models (Wagner et al., 2019).

Nevertheless, in all these studies, state-estimation has been considered independently for water or energy systems, and the interlinked

nature of water-energy networks has been out-of-scope. The main challenge of independent state-estimation for water or energy network is that it treats the other counterpart as a black box and fails to fully observe the interlinked system as a whole to protect it from interruptions or threats. For example, if FDIs occur in one infrastructure, they may not be detected in the counterpart network, causing damage to the core operation of the nexus. Several studies focused on modeling and detection of FDIs in water or energy networks. For instance, the Environmental Protection Agency Network (EPANET) (the name of a hydraulic simulation software) and a machine-learning algorithm were used to detect cyber-physical attacks in WDS in Ramotsoela et al., 2019. In another study, EPANET and mixed integer linear programming (MILP) were used to identify cyber attacks in a looped WDS (Housh and Ohar, 2018). Some studies considered gravitational-based WDSs, which cannot be implemented in the looped pressurized systems (Amin et al., 2012; Amin et al., 2012). One study developed a simulation toolbox, i.e., epanetCPA, to emulate a physical- and cyber-components of WDSs (Taormina et al., 2017). However, none of these studies considered the energy network in the state-estimation or detection process.

In power systems, however, bad data detection has been well studied (Kosut et al., 2011; Teixeira et al., 2010; Eder-Neuhauser et al., 2017; Ganjkhanian et al., 2019). Several studies have identified cyber attacks with mathematical models (Liu et al., 2011; Hug and Giampapa, 2012; Giani et al., 2013), while others have developed detection algorithms by enumerating the attack vectors (Dan and Sandberg, 2010; Teixeira et al., 2010). However, the studies on bad data detection of energy systems consider WDSs as one of the loads, mixed with other loads, and thereby treat it as a measurement unit. As such, any FDI on the load terminal will be treated as false data readings on power consumption of the water distribution system. However, in reality, the false data readings on the water distribution terminal could stem from FDIs in a variety of readings such as nodal pressure, tank level, or the pump's speed. Accordingly, only an integrated state-estimation and bad data detection system can precisely detect the origin of bad data in a combined water-energy network, and this motivates the contributions of this research.

There has been a limited number of efforts in detecting bad data in integrated water-energy systems. For example, optimization methods were developed to enhance the resilience of an integrated water-energy system in (Zuloaga et al., 2019). However, these models were specifically developed for limited water and/or energy availability scenarios, while other potential anomalies, such as FDIs in the pressure head of the tank or possible attacks on flow of the pumps or reservoir, were not considered. Moreover, state-estimation was not considered in energy nor in water distribution network. Another study developed a sub-observer-based fusion filter to detect bad data in a combined water-energy system inside a power plant (Khalid et al., 2019). In this system, the interdependence of water and energy was defined within the power plants in the form of cooling water basins, desalination unit, and condensers. Hence, the energy system and the water distribution system were disconnected, and water distribution system was not within the scope of the study. Furthermore, the energy network was limited to a power plant only, and the state-estimation of an energy network was not considered. Therefore, to the best of the authors' knowledge, prior research did not focus on developing a comprehensive framework for state-estimation and bad data detection of integrated water-energy systems.

1.2. Contribution

To address the limitations of the existing research including detailed mathematical state-estimation and bad data detection models that contemplate both water and energy equations, as well as avoiding the computational difficulties that prior research reported in terms of developing the water state-estimation due to the existence of the non-linear equations, this paper proposes an integrated state-estimation and

bad data detection framework for a combined water-energy system such that.

- The state-estimation of water distribution network is designed as a three-tier state-estimation approach that improves computational challenges due to nonlinearity of equations involved in estimating the pressure head at the nodes and the flows in the pipes. Specifically, the proposed three-tier iteration method is an alternative to the iteration-simulation method developed previously (Arsene and Gabrys, 2014), offering an expedited and more straightforward state-estimation for the water system with similar backbone to the energy state-estimation framework, which will facilitate the convergence of the two models into a single conjunction,
- An energy state estimator is designed to provide the estimation of parameters of the energy system (including bus voltages, power injections and transmission flows) in parallel to the water state-estimation,
- The interlink between water and energy state-estimation is modeled by using the estimated parameters of the water distribution system as measurements in the state-estimation process of the energy system considering load dynamics,
- The outcomes of the integrated water-energy state estimator is sent to an “Integrated Bad Data Detection Center” (IBDDC) to identify any potential bad data and take corrective actions.

An integrated energy-water system composed of IEEE (stands for Institute of Electrical and Electronics Engineers) 14-bus power distribution network and a 12-node water distribution system is developed for the case studies using MATLAB software. Several case studies are run to analyze the performance of the proposed bad data detection framework on detecting bad data on water distribution system or energy network. The rest of the paper is organized in the following order. The state-estimation model of the water sector is formulated in Section 2, the state-estimation of the energy network is modeled in Section 3. Case studies are included in Section 4 and Section 5 concludes the paper.

2. Problem formulation

Given the topology and parameters of the system, the state-estimation requires the readings from the measurement devices (including phasor measurement units (PMUs), remote terminal units (RTUs), or smart meters) in the integrated water-energy network and is started once all the measurements have been received. The framework for integrated state-estimation will be implemented in supervisory control and data acquisition (SCADA) center of the integrated system and is one of the main functions of the SCADA center. The structure of the system (such as topology of the system, parameters of the lines, size of the loads and generation units) is known and cannot change during the estimation process.

An overview of the proposed framework is demonstrated in Fig. 1,

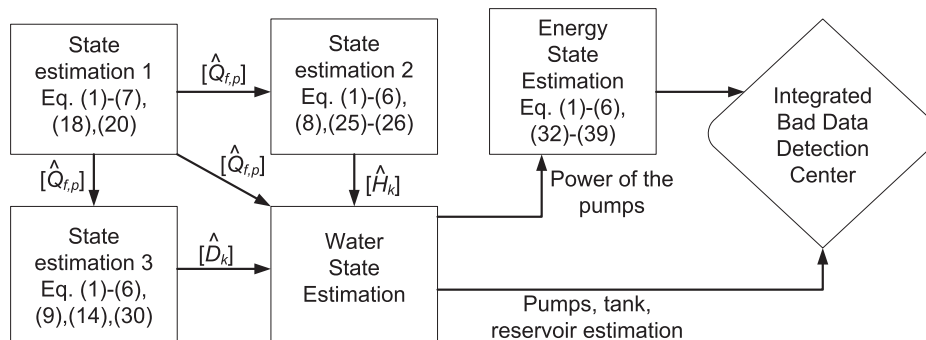


Fig. 1. Water-energy state-estimation flowchart.

in which a 3-tier state-estimation of the water system and one-step estimation of energy network parameters are designed to simultaneously process measurement inconsistencies in the water-energy system. One of the challenges in developing the state-estimation method for water systems is the nonlinear correlations between the nodal pressure head, head-loss of the pipe, and pump's head gain with the corresponding flow rate, expressed in Bernoulli, Hazen-Williams, and pump's characteristic equations, respectively. These nonlinear equations generate singular Jacobian matrix, especially when flow rate is small in some of the pipes. To resolve this issue, previous researchers tried to obtain the nodal heads (associated with the nonlinear equations) from other methods such as simulations (Arsene and Gabrys, 2014) and then combined them with iterative methods such as Newton-Raphson to acquire other state variables like nodal demands, which can be derived from the linear mass continuity theorem (14). In the present work, a coherent state-estimation framework for both water and energy systems is designed to facilitate the merge of the water and energy systems state-estimation algorithms. For this reason, the water system state-estimation is designed in three steps of Newton-Raphson, each step consisting of linear equations, to obtain all state variables within one single consistent model without the need for contemplating a simulation model. First, loop corrective flows are estimated in the state-estimation 1 block by calculating the flow rates of the pipes ($[Q_{f,p}]$) in the system. These in-pipe flows will serve as input values for estimation of both the heads ($[H_k]$) and the demands of the nodes ($[D_k]$) through state-estimation 2 and 3, respectively. The in-pipe flows will also be used to estimate the power consumption of the pumps. The power demands, associated with the pumps, will act as input loads for the energy state estimator. The estimation outcomes of both the water and energy state-estimation will be sent to the integrated bad data center for false data injection processing. Details of each state-estimation process will be elaborated in the next sections.

3. State-estimation of the water system

In the water sector, most of the water utility companies use telemetry systems, as an alternative method to transmitting data, which are employed for obtaining data on the present state of the network and measuring flows and pressures (Tshehla et al., 2017; Shende and Chau, 2019). However, the large scale of WDSs does not allow for all the pipe flows, nodal pressures, and nodal demands to be measured. Hence, a mathematical model of the water system consisting of mass and energy balance equations in the network's nodes and links is used. Next, a maximum likelihood estimation (MLE) technique in least square estimation (LSE) process is implemented to estimate the parameters of the water network (Gould et al., 2006). The objective function is designed to minimize the summation of measurement residuals

$$\min_{x_i} J(x) = \sum_{i \in N_m} \frac{[Z_i - g_i(x_i)]^2}{\sigma_i^2} \quad (1)$$

where N_m is the number of measurements, Z_i are the measurements or pseudo-measurements of the water system variables, including the flow rate of the pumps, as well as the head and flow of the tank and reservoir. In this equation, x_i are the state variables (nodal heads and demands), and $g_i(x_i)$ are the hydraulic equations related to the state variables of the water system. Through the Newton–Raphson technique, which is a well-known method (Gould et al., 2006), the gradient of measurement residuals ($\nabla J(x)$) is forced to zero, as the minimum amount of the function $J(x)$ is obtained when $\frac{\partial J}{\partial x_i} = 0$. Hence

$$\nabla J(x) = 0 \rightarrow \left[\frac{\partial J}{\partial x_1}, \dots, \frac{\partial J}{\partial x_2}, \dots, \frac{\partial J}{\partial x_N} \right]^T = 0 \quad (2)$$

By substituting $J(x)$ from (1) into (2), the estimated value of x can be iteratively solved until the convergence (very small threshold) is achieved

$$[\hat{x}]^{i+1} = [\hat{x}]^i + \Delta[x], \quad (3)$$

$$\Delta[x] = [H]^T [R]^{-1} [H]^T [R]^{-1} \begin{bmatrix} Z_1 - g_1(x) \\ Z_2 - g_2(x) \\ \vdots \end{bmatrix} \quad (4)$$

where i is the iteration number, $[\hat{x}] = [\Delta\hat{Q}_i, \hat{H}_k, \hat{D}_k]$ is the estimated state vector and $[H]$ and $[R]$ are the Jacobian and covariance matrices, respectively, and are represented as

$$[H] = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_1}{\partial x_N} \\ \frac{\partial g_2}{\partial x_1} & \dots & \frac{\partial g_2}{\partial x_N} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_N}{\partial x_1} & \dots & \frac{\partial g_N}{\partial x_N} \end{bmatrix} \quad (5)$$

and

$$[R] = \begin{bmatrix} \sigma_1^2 & 0 & 0 & \dots \\ 0 & \sigma_2^2 & 0 & \dots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \sigma_N^2 \end{bmatrix} \quad (6)$$

In (4), $g_i(x)$ is expressed in (7)–(9).

As shown in Fig. 2, three individual Newton–Raphson (N-R) iterative calculations are developed to obtain the state variables, which is explained in detail in the next section. The $g_i(x_i)$ for each of these N-R calculations are represented as

$$[g(\Delta Q_i)] = [M_{lp}][h_{loss,p}] \quad (7)$$

$$[g(H_k)] = [H_{k-1} - H_k] \quad (8)$$

$$[g(D_k)] = [D_k] \quad (9)$$

where in (7), ΔQ_i is the loop corrective flow, $h_{loss,p}$ is the head loss of the pipe p , expressed as (Houghtalen et al., 2016)

$$h_{loss,p} = F_p \times (\Delta Q_i)^{1.85} \quad (10)$$

where F_p is the Hazen–Williams coefficient calculated as

$$F_p = \frac{L_p}{(0.278 C_p D_p)^{1.85}}$$

in which, L_p , D_p , and C_p are the length (m), diameter (m), and roughness coefficient of the pipe p , respectively. All the parameters used in this study are provided in Table A1, in the appendix.

Matrix M_{lp} is the loop incidence matrix as below

$$M_{lp} = \begin{bmatrix} M_{11} & M_{12} & \dots & M_{1p} \\ M_{21} & M_{22} & \dots & M_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ M_{l1} & M_{l2} & \dots & M_{lp} \end{bmatrix} \quad (11)$$

such that

$$M_{m,n} = \begin{cases} 1 & \text{direction of flow } n \text{ and loop } m \text{ are the same} \\ -1 & \text{direction of flow } n \text{ and loop } m \text{ are opposite} \\ 0 & \text{flow of } m \text{ does not belong to loop } n \end{cases}$$

In (8), H_{k-1} and H_k are the heads at consecutive nodes $k-1$ and k , obtained from Bernoulli's equation (Houghtalen et al., 2016)

$$H_{k-1} - H_k - h_{loss,p} = \Delta h_j \quad (12)$$

where Δh_j is the head gain of the pump (Walski et al., 2003)

$$\Delta h_{j,l} = a_j q_{pj}^A + b_j q_{pj}^B + c_j \quad (13)$$

where q_{pj} is the flow measurement of pump j obtained from SCADA readings, and a_j , b_j , c_j , A , and B are parameters of the pump's characteristics equation (provided in Table A1).

In (9), the nodal demands are calculated from mass balance equation at each node (Houghtalen et al., 2016)

$$[D_k] = [A_{kp}][Q_p] \quad (14)$$

where matrix $[A_{kp}]_{k \times p}$ is the network incidence matrix for k -node and p -pipe obtained as

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1p} \\ A_{21} & A_{22} & \dots & A_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ A_{k1} & A_{k2} & \dots & A_{kp} \end{bmatrix} \quad (15)$$

and

$$A_{i,j} = \begin{cases} 1 & \text{flow of pipe } p \text{ entering node } k \\ -1 & \text{flow of pipe } p \text{ leaving node } k \\ 0 & \text{flow of pipe } p \text{ is not related to node } k \end{cases}$$

and $[Q_p]$ is the matrix of pipe flows which will be obtained from N-R1. These functions will be used in the three N-R iterations of water estimation, explained in detail in the following subsections.

3.1. N-R1: Loop corrective flows as state variables

As demonstrated in Fig. 2, the first Newton–Raphson calculations (N-R1) are developed to acquire loop corrective flows (ΔQ_i), to be used to calculate the flows of the pipes ($Q_{f,p}$). This is included because when nonlinear Hazen–Williams equation for head-loss in nodal head estimation is applied, N-R method has been shown to have computational issues (Bargiela, 1984). In that case, the inverse of the Jacobian matrix approaches singularity when flows in the pipes are small. Hence, N-R1 is incorporated to solve for loop corrective flows, representing the flow of each loop independent from the pipes, to compute total head loss of the loop. This is based on the conservation of energy theorem, which states that the difference between the energy head of any two points in a WDS is equal to the energy gained (via pump) minus the energy lost through head-loss between the two points (Houghtalen et al., 2016). As such, in a loop where the initial and final node is the same, the summation of the head-loss of the pipes belonging to the loop is equal to zero, unless there exists a pump in which case the summation of the head-losses is equal to the energy head of the pump, expressed in (17). Loop corrective flows estimation is essential to obtaining state variables of nodal heads and nodal demands because of the following. The water system's state variables are functions of pipes' flow rates through nonlinear Bernoulli's Eq. (12) and linear mass continuity theorem (14), respectively. However, flow rate measurements are only available for the main arteries and the pipes with pumps, due to the economic burdens that installation of smart meters on every pipe can impose on water systems. Herein, loop corrective flows will allow for a primary estimation of nodal heads, in which flow rates are replaced with loop corrective flows in calculations. The outcome of the N-R1 will be used

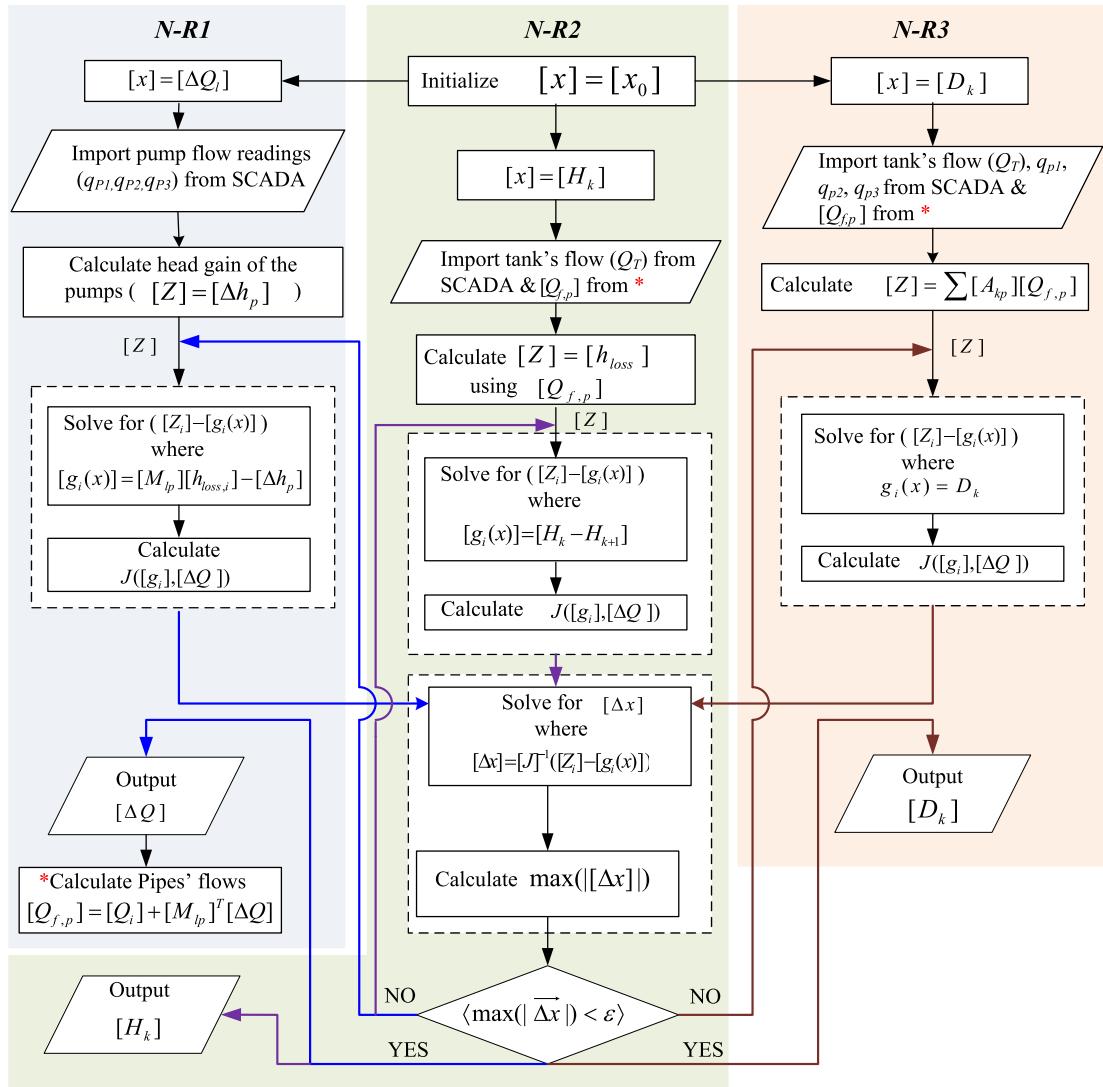


Fig. 2. Water system's state-estimation flowchart.

for nodal head calculations via linear equations (Arsene and Gabrys, 2014).

As shown in Fig. 3, the studied network consists of 7 loops, which consequently generate seven corrective flows shown by vector ΔQ_l , where l is the loop identification in the Fig. 2, e.g. ΔQ_1 belongs to loop 1.

$$\Delta Q_l = [\Delta Q_1 \ \Delta Q_2 \ \Delta Q_3 \ \Delta Q_4 \ \Delta Q_5 \ \Delta Q_6 \ \Delta Q_7]^T$$

For any loop l and any pipe p connecting nodes k and $k - 1$, including a pump j , Bernoulli's equation is defined as (12). Plugging in M_{lp} , nodal heads, head losses of the pipes, and head gains of the pumps into (12), the energy equation of the loops are provided as

$$[\Delta H_l]_{1 \times 1} = [M_{lp}]_{1 \times p} [h_{loss,p}]_{p \times 1} \quad (16)$$

where

$$\Delta H_l = \begin{cases} \Delta h_{lj} & \text{pump } j \text{ is in the loop } l \\ 0 & \text{otherwise} \end{cases} \quad (17)$$

and Δh_{lj} is the head gain of the pump j in loop l , obtained from (13). The maximum likelihood criterion, derived from (1), is used to obtain $[x] = [\Delta Q_l]$. In (1), $[Z_i] = [\Delta H_l]$, which consists of residual loop head losses expressed in (16). Hence, the objective function in the N-R1 is

$$\min_{|\Delta Q_l|} J \left(\left| \Delta Q_l \right| \right) = \sum_{i \in N_l} \frac{[\Delta H_l - g_i(\Delta Q_l)]^2}{\sigma_i^2} \quad (18)$$

Through the Newton-Raphson technique (2), the gradient of measurement residuals ($J(\Delta Q_l)$) is forced to zero. By substituting $J(\Delta Q_l)$ from (18) into (2), the estimated value of the loop corrective flow at step $i + 1$ can be derived from (3) such that

$$[\hat{x}] = [\hat{\Delta Q}_l] \quad (19)$$

where $\Delta[x] = \Delta[\Delta Q_l]$ is derived from (4), $[H]$, and $[R]$ are derived from (5) and (6), respectively. Using the loop corrective flows derived from the first Newton-Raphson loop, the final flows in each pipe, $\hat{Q}_{f,p}$, can be found using

$$[\hat{Q}_{f,p}] = [Q_{i,p}] + [M_{lp}]^T [\hat{\Delta Q}_l] \quad (20)$$

where $\hat{\Delta Q}_l$ are the loop corrective flows, $Q_{i,p}$ are the initial flows in the pipes, among which $Q_{i,1} = Q_{i,T}$ is the inflow from the tank, $Q_{i,3} = q_{p2}$ is the flow of pump 2, $Q_{i,7} = q_{p3}$ is the flow of pump 3, and $Q_{i,11} = q_{p1}$ is the flow of pump 1 are measured and the rest are "guessed" based on these given flows and simple splits among the branches of the network. These initial guessed values will be corrected for each pipe once $\hat{\Delta Q}_l$ are applied in (20). Matrix $[M_{lp}]^T$ is the transpose of matrix $[M_{lp}]$ in (11). Next, the head loss of the pipes are adjusted by replacing ΔQ_l and $q_{p,j}$ (for the

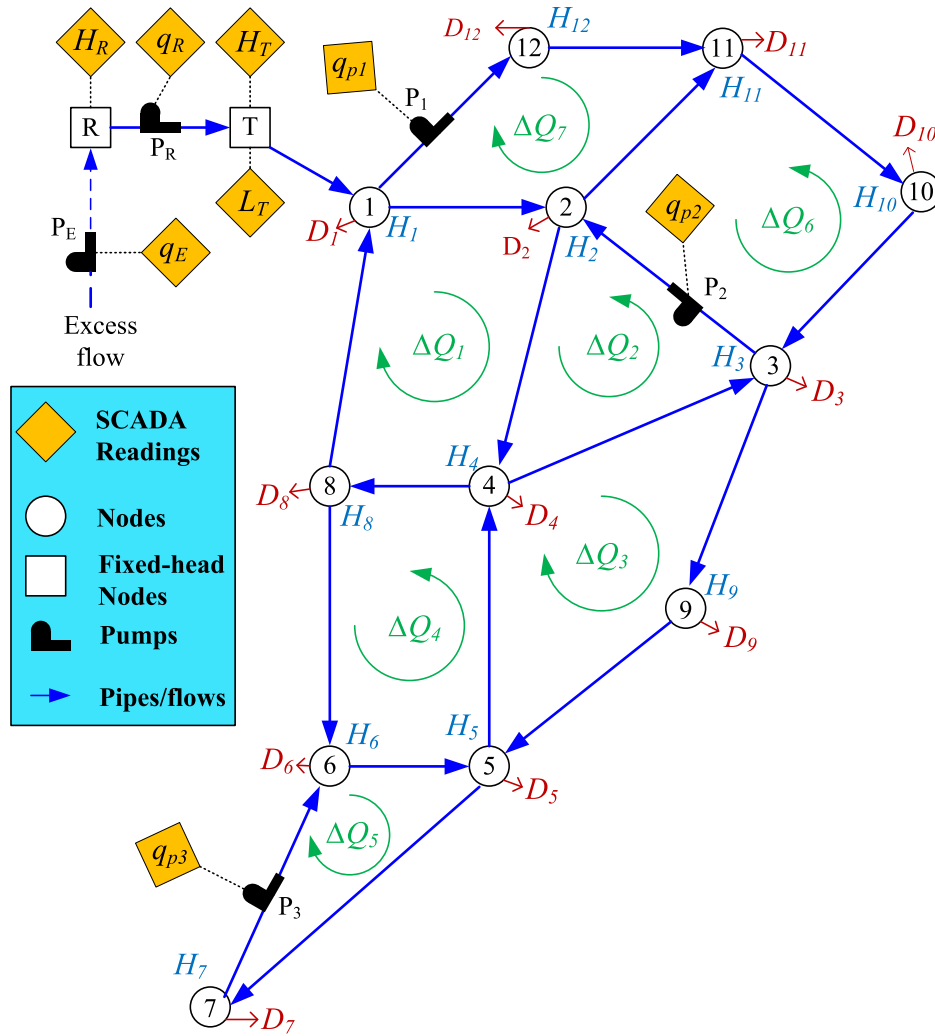


Fig. 3. Studied water network.

pipes including the pumps) with corresponding $\hat{Q}_{f,p}$ in (10) and (13). Hence

$$h_{loss,p}^* = F_p \times (\hat{Q}_{f,p})^{1.85} \quad (21)$$

$$\Delta h_j^* = a_j (\hat{Q}_{f,pj})^A + b_j (\hat{Q}_{f,pj})^B + c \quad (22)$$

As shown in Fig. 2, the results will be served as pseudo-measurements in the next N-R calculations. Additionally, $q_{p,j}$ and $\hat{Q}_{f,pj}$ will be used to calculate the “measured” and “estimated” power consumption of the pumps, respectively, to be sent to the energy state-estimation, expressed as (Walski et al., 2003)

$$P_{m,j} = \frac{\gamma \times \Delta h_j \times q_{p,j}}{\eta} \quad (23)$$

$$P_{es,j} = \frac{\gamma \times \Delta h_j^* \times \hat{Q}_{f,pj}}{\eta} \quad (24)$$

where $P_{m,j}$ and $P_{es,j}$ are the measured and estimated power demand for pump j , γ is the specific weight of water, which is 9.789 kN/m³ at 20°C, and η is the pump's efficiency, assumed to be 80% for all pumps (Ulanicki et al., 2008).

3.2. N-R2: Nodal heads as state variables

As illustrated in Fig. 2, the second iterative calculations, i.e., N-R2, will take in $\hat{Q}_{f,p}$ from N-R1 to calculate the nodal heads considering

function $g(H_k)$ as (8). From (12), the matrix of measurements for this step, $[Z]$, is derived as

$$[Z] = [h_{loss,p}^*] - [\Delta h_j^*] \quad (25)$$

Three remarks should be noted here: (1) As exhibited in Fig. 3, there are 12 nodes and 19 head losses in the studied water network, meaning that only 12 out of 19 should be selected to include in the measurements. As such, those pipes with actual SCADA measurements including pipe 1 (connecting the water tank to node 1) and the pipes containing q_{p1} , q_{p2} , and q_{p3} (i.e., the pumps' flows) are contemplated, while the rest are chosen in a way that the inverse of the Jacobian matrix will not become singular; (2) for those pipes without a pump, Δh_j^* is considered zero in the matrix $[\Delta h_j^*]$; and (3) the head of the tank, H_T , is given as one of the SCADA readings.

Hence, from (1), the objective function in the N-R2 is

$$\min_{|H_k|} J \left(\left| H_k \right| \right) = \sum_{i \in N_k} \frac{[Z - g_i(H_k)]^2}{\sigma_i^2} \quad (26)$$

where N_k is the number of nodal head measurements. From N-R method in (2), the gradient of measurement residuals ($J(H_k)$) from (2) is forced to become zero. By substituting $J(|H_k|)$ from (26) into (2), nodal heads can be estimated from (3) and (4) such that

$$[\hat{x}] = [|\hat{H}_k|] \quad (27)$$

where in (4), $[H]$ is calculated from (5) and $[R]$ is obtained from (6). The

outcome of N-R2 is a 12×1 matrix with row entries equal to the head of the corresponding node.

3.3. N-R3: Nodal demands as state variables

As it is demonstrated in Fig. 2, in the third column from left, nodal demands are calculated using the pipes' flows estimation from N-R1 (20), while the flow of those pipes with pumps are entered from the pumps' flow measurements, i.e., q_{p1} , q_{p2} , and q_{p3} , given by SCADA. Hence, the matrix of measurements, $[Z]$, is found by applying the $[Q_{f,p}]$ in (14), where

$$[Q_{f,p}] = \begin{cases} \hat{Q}_{f,p} & \text{pipes without pump from (20)} \\ q_{p,j} & \text{pipes with pump } j \text{ from SCADA} \end{cases} \quad (28)$$

Thus,

$$[Z] = [A_{kp}][Q_{f,p}] \quad (29)$$

The water demand at each node can be calculated by solving the objective function from (1), which is modified for nodal demands as

$$\min_{|D_k|} J \left(\left| D_k \right| \right) = \sum_{i \in N_k} \frac{[Z - g_i(D_k)]^2}{\sigma_i^2} \quad (30)$$

where N_k is the number of nodal demand measurements and $g_i(D_k)$ is substituted from (9). By performing N-R3 method, the gradient of measurement residuals ($J(D_k)$) from (2) is forced to become zero, in which $J(|D_k|)$ is replaced from (30). The nodal demands at step $i + 1$ can be obtained from (3) and (4) such that

$$[\hat{x}] = [|\hat{D}_k|], \quad (31)$$

where in (4), the Jacobian matrix $[H]$ is obtained from (5), and $[R]$ is from (6). The result of N-R3 is a 12×1 matrix with row entries equal to demand at each node.

In the next section, state-estimation method of the energy sector is explained.

4. State-estimation of energy network

The maximum likelihood criterion is also used in least square estimation process to estimate the parameters of the energy network. The objective is to minimize the sum of measurement residuals from (1), in which, Z_i are energy system measurements at nodes (buses) including voltage magnitude and angle ($|V_i| \angle \theta_i$), active power injection to the bus (P_i), reactive power injection at nodes (Q_i), active power flow from node i to j (P_{ij}) and reactive power flow from node i to j (Q_{ij}). In (1), x_i is the state variable to be estimated (voltage magnitudes and angles at different nodes), $g_i(x_i)$ are nonlinear functions relating the state variables to measurement values, N_m is the size of measurement vector, σ_i^2 is the measurement error for the i th measurement, and $J(x)$ is the measurement residual. The $g_i(x_i)$ functions in an energy system can be represented by (Wood et al., 2013)

$$P_k = |V_k| \sum_{n=1}^N |V_n| [G_{kn} \cos(\theta_k - \theta_n) + B_{kn} \sin(\theta_k - \theta_n)] \quad (32)$$

$$Q_k = |V_k| \sum_{n=1}^N |V_n| [G_{kn} \sin(\theta_k - \theta_n) - B_{kn} \cos(\theta_k - \theta_n)] \quad (33)$$

$$P_{ij} = G_{ij} |V_i|^2 - |V_i| |V_j| [G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)] \quad (34)$$

$$Q_{ij} = -B_{ij} |V_i|^2 - |V_i| |V_j| [G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j)] \quad (35)$$

where G_{ij} and B_{ij} are real and imaginary components of the bus admittance matrix ($[Y_{bus}]$), respectively. The bus admittance matrix is an $N \times N$ matrix whose ij th element depends on the admittance of the lines connected between nodes i and j in the energy system and is found

by

$$Y_{bus} \begin{pmatrix} i, j \end{pmatrix} = \begin{cases} \sum \text{admittances connected to bus } i, & \text{if } i = j \\ - \sum \text{admittances connected between bus } i \text{ and } j, & \text{if } i \neq j \end{cases}$$

It is also noted that if the measurements are voltage magnitudes or angles, Eq. (1) can be simplified to

$$\min_{|V_i|} J(|V|) = \sum_{i \in N_V} \frac{[|V_i^m| - |\hat{V}_i|]^2}{\sigma_{V_i}^2} \quad \min_{\theta_i} J(\theta) = \sum_{i \in N_\theta} \frac{[|\theta_i^m| - |\hat{\theta}_i|]^2}{\sigma_{\theta_i}^2} \quad (36)$$

where N_V is the number of voltage magnitude measurements, N_θ is the number of voltage angle measurements, $\hat{V}_i$ and $|\hat{\theta}_i|$ are estimated voltage magnitude and angle at bus i , and $|V_i^m|$ and θ_i^m are measured voltage magnitude and angle at bus i . Since the relationship between the state variables ($|V_i|$, θ_i) and some measurements such as active and reactive power injections/flows (Eqs. (32)–(35)) is not linear, an iterative technique needs to be carried out to solve the state-estimation problem. Through Newton–Raphson method in (2), the gradient of measurement residuals ($J(x)$) are forced to zero.

$$\nabla J \left(\left| V_i \right|, \theta_i \right) = 0 \rightarrow \left[\frac{\partial J}{\partial \theta_1}, \dots, \frac{\partial J}{\partial \theta_N}, \frac{\partial J}{\partial |V_1|}, \dots, \frac{\partial J}{\partial |V_N|} \right]^T = 0 \quad (37)$$

By replacing $J(|V_i|, \theta_i)$ from (36) (if the measurement is voltage magnitude or angle) or (1) (if the measurement is not voltage magnitude nor angle) in (37), the estimated value can be derived using Newton's iterative procedure in (3) and (4) such that

$$[\hat{x}] = \left[\frac{\theta}{|V|} \right] = [\hat{\theta}_1, \hat{\theta}_2, \dots, |\hat{V}_1|, |\hat{V}_2|, \dots]^T \quad (38)$$

and depending on the measurement type, $g_i \left(\left[\frac{\theta}{|V|} \right] \right)$ can be to the form of (32)–(35), or $|V_i|$, θ_i as direct voltage measurements in (36). The covariance matrix, $[R]$, is calculated by (6) and the Jacobian matrices, $[H]$, is derived from (5) represented by

$$[H] = \begin{bmatrix} \frac{\partial f_1}{\partial \theta_1} & \dots & \frac{\partial f_1}{\partial \theta_N} & \frac{\partial f_1}{\partial |V_1|} & \dots & \frac{\partial f_1}{\partial |V_N|} \\ \frac{\partial f_2}{\partial \theta_1} & \dots & \frac{\partial f_2}{\partial \theta_N} & \frac{\partial f_2}{\partial |V_1|} & \dots & \frac{\partial f_2}{\partial |V_N|} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_N}{\partial \theta_1} & \dots & \frac{\partial f_N}{\partial \theta_N} & \frac{\partial f_N}{\partial |V_1|} & \dots & \frac{\partial f_N}{\partial |V_N|} \end{bmatrix} \quad (39)$$

Derivation of Jacobean matrix elements ($[H]$) is well studied in power system textbooks and therefore, is skipped in this section. Readers are encouraged to refer to (Wood et al., 2013) for detailed derivation and procedure for the state-estimation of energy networks. The flowchart of the proposed integrated state-estimation and bad data detection method for water-energy systems is illustrated in Fig. 4. As it can be seen, the energy network state-estimation is conducted on the right side of the flowchart, where measurements from the energy network and readings of the pumps from water state-estimation will be used to solve for the Newton–Raphson loop and calculate the estimated values of voltages, power injections, and power flows. The Integrated Bad Data Detection Center (IBDDC) receives estimated values from integrated state-estimation procedure and compares the estimated values with measurements to detect any bad data that exists in water or energy network.

5. Case studies

Three case studies are included to investigate the impact of implementing an integrated bad data detection center on the operational decision and security in a combined water-energy network, and compare those with independent traditional bad data detection systems. A schematic description of the studied system is exhibited in Fig. 5, in which a standard IEEE 14-bus benchmark system is combined with a

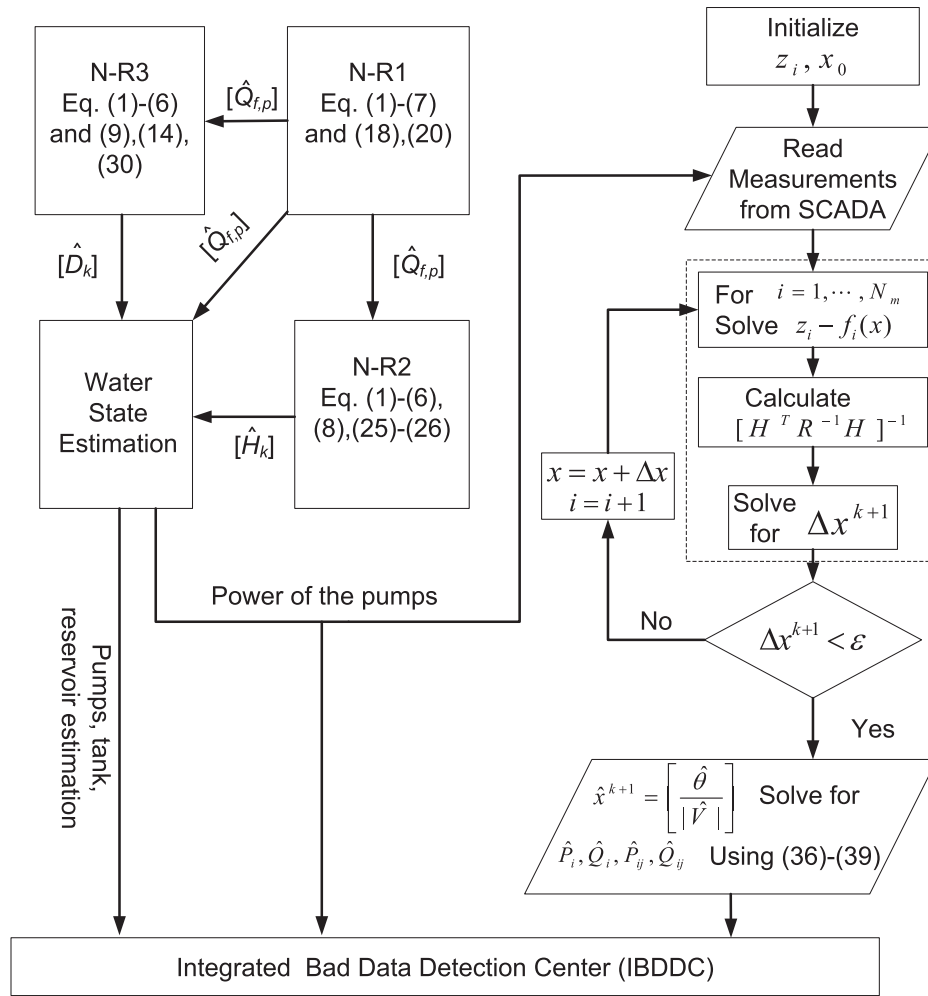


Fig. 4. Integrated water-energy state-estimation flowchart.

water distribution network benchmark known as “Battle of the Water Sensor Networks”. The energy network resembles the U.S. power grid in 1962 and the water distribution network resembles an area in Kentucky, U.S. in 2014. The water network consists of one reservoir, one water tank, 5 pumps, and 12 nodes. The entire model was developed in MATLAB software, where the parameters of the IEEE 14-bus system were adopted from an open-source “MATPOWER” toolbox (Zimmerman et al., 2010) and parameters of the water distribution benchmark are listed in Table A1, obtained from (Hernandez et al., 2016). While buses No. 2 and 3 supply electricity to the residential buildings, bus No. 3 supplies power to the pumps P_R , P_2 , and P_E and bus No. 2 provides power to the other two pumps of P_1 and P_3 . The power of the pumps are calculated from (23) and (24). As shown in Fig. 3, pump P_R transfers water from the reservoir to the tank, located at a higher elevation, and is one of the most critical elements of the water system. Pumps P_1 , P_2 , and P_3 are located on the residential pipes to compensate low pressure in the associated nodes, but are not considered vital to the network, since the grid design of the water system allows the water utility to send water from other branches if these pumps are shut off. Finally, the pump P_E is installed on a main pipe returning the excess water from the residential network back to the reservoir. Four case studies are included to validate the proposed model. The first case study examines and discusses the integrated state-estimation model in the absence of bad data, generating base-lines for the unknown parameters and maximum estimation errors for the water and energy systems. In the next three case studies, the performance of the standalone and integrated models are discussed under various circumstances; (1) FDI in the flow rate of pump 2, (2) FDI in the flow rate of pump R, and (3) FDI

in the level of the water in the tank. These scenarios represent failures in less critical (e.g., pump 2’s flow rate), critical (e.g., reservoir effluent), and overly critical (e.g., tank’s water level) components of the water system, and are carefully designed to study the effect of these fault events on the combined water-energy system state-estimation and the integrated bad data detection algorithms.

5.1. State-estimation of integrated system with no bad data

The proposed integrated state-estimation was run for the system illustrated in Fig. 5 to estimate the parameters of the integrated network. Table 1 presents the nodal heads ($\hat{H}_k$), nodal demands ($\hat{D}_k$), and the pipes’ flows ($\hat{Q}_{f,p}$) in the water distribution network, attained from the proposed three-tier Newton-Raphson estimation technique. Additionally, Table 1 includes the voltage magnitude ($|V_i|$) and angle (θ_i), active and reactive power injections (P_i and Q_i), and active and reactive power flows ($P_{i,j}$ and $Q_{i,j}$) of the studied electric power network. The data associated with the fixed-head nodes, i.e. water tank and reservoir, and pumps are not included as such information is provided in Table A1. An error tolerance of 0.0001 is considered for all the estimated values in the integrated water-energy network (Abur and Exposito, 2004). Results in Table 1 confirm that the state-estimation can successfully estimate the parameters of the integrated network.

5.2. False data injection on flow of pump 2 (q_{p2})

A random false data equal to 18% of the nominal value of pump 2

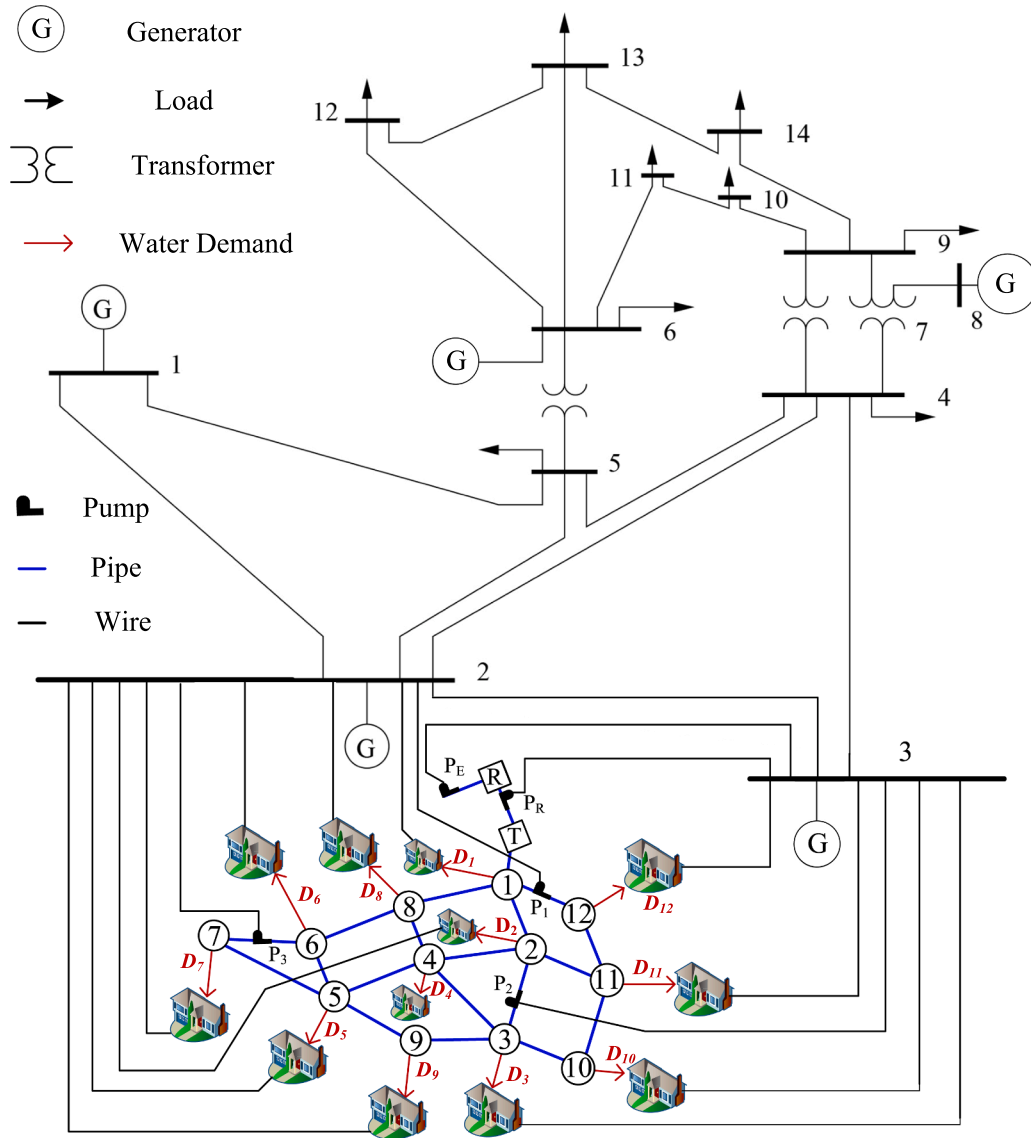


Fig. 5. Modified IEEE 14-bus system combined with a 12-node water distribution system.

flow rate is injected to the measurements of pump 2 to examine how the standalone energy state-estimation and the integrated state-estimation detect potential bad data. The selected FDI is small to not be detected by the SCADA system but large enough to cause failure in the system. The left subplot in Fig. 6 demonstrates all the estimated data by the energy state estimator, showing clean data by blue circles, while the bad data detected are shown with red cross. The energy and water tolerance thresholds are the maximum mean squared deviation (MMSD) of the estimated parameters with their measurements associated with the energy and water systems in the absence of bad data, respectively. In other words, the systems are calibrated such that if the average squared difference between the estimation and measurement values of a parameter is less than or equal to MMSD, that parameter is considered to be “clean”. The MMSD of the studied energy and water systems are evaluated to be 0.9% and 10%, respectively. In this case, the error between the estimated and measured values of the active power injection of bus No.3 (P_3) and the active power flow ($P_{2,3}$) were calculated to be higher than the threshold, i.e., 0.9% in the energy system, thus, were flagged as bad data by the energy state estimator. However, the detection system cannot process the origin of the FDI in the system. On the other hand, the integrated detection system, shown in the right subplot in Fig. 6, presents the estimated data of the

integrated system, where the horizontal green line marks the water system threshold as 10% error between the estimated and measured data. The green circles denote the clean estimated measurements and green rectangles represent the bad data in the water distribution network, meaning the green rectangles reveal the parameters for which the mean squared deviation of the estimated and measured values are higher than the calibrated error of 10% in the water system. In this subplot, the water and energy data associate with the right and left y-axes, respectively. As can be seen in the left subplot, the IBDDC flagged 4 data points as bad data, which indicates bus No.3 is compromised and identifies the origin of this FDI to be pump 2 in the water distribution system. This is a compelling finding as it crafts the systems management in the time of crisis due to the following. Bus No.3 supplies power to pump 2, which is why it is compromised by FDI in the pump, but it also provides the electricity of the homes in this network. The operational decision as an outcome of the standalone energy system estimation (the left subplot) may result in disconnecting all the loads on bus No.3, and thereby, a power outage will be experienced in parts of the neighbourhood. However, the proposed integrated state-estimation (the right subplot) results in a shutdown for pump 2, while the rest of the system will operate without any problem. This clearly demonstrates the advantage of an integrated state estimation for water-energy networks,

Table 1
State Variables of the Water and Energy Network.

Node No.	H_k m	D_k LPS	from-to Node	$Q_{f,p}$ $\times 10^2$ LPS	from-to Node	$Q_{f,p}$ $\times 10^2$ LPS
1	58.30	254.85	T-1	3.35	10-3	0.349
2	58.30	16.93	1-2	0.0007	9-5	0.353
3	44.18	22.85	3-2	0.536	8-6	0.349
4	34.92	0.00	4-3	0.300	12-11	0.349
5	27.24	0.85	5-4	0.298	2-4	0.351
6	25.61	30.44	6/5	0.308	8-1	0.350
7	108.78	8.71	7/6	0.271	5-7	0.200
8	38.30	20.07	3/9	0.349		
9	11.42	0.00	11/10	0.353		
10	43.56	0.00	2/11	0.353		
11	43.24	35.00	1/12	0.352		
12	101.95	64.92	4/8	1.00		

Bus NO.	$ V $ p.u.	θ_i Degree	P_i p.u.	Q_i p.u.	from-to Bus NO.	P_{ij} p.u.	Q_{ij} p.u.
1	0.972	0.00	2.35	-0.095	1-2	1.58	-0.171
2	0.955	-5.95	0.119	0.357	1-5	0.773	0.0754
3	0.924	-13.85	-0.692	0.074	2-3	0.623	0.0255
4	0.919	-12.55	-0.651	0.038	2-4	0.584	0.019
5	0.923	-10.735	-0.098	-0.044	2-5	0.439	0.034
6	0.983	-17.588	-0.109	0.220	3-4	-0.089	0.055
7	0.954	-16.14	0.0002	0.010	4-5	-0.601	0.117
8	0.993	-16.14	0.0002	0.218	4-7	0.27	-0.150
9	0.939	-18.033	-0.271	-0.148	4-9	0.15	-0.026
10	0.937	-18.41	-0.117	-0.058	5-6	0.461	-0.207
11	0.956	-18.12	-0.032	-0.017	6-11	0.087	0.090
12	0.964	-18.60	-0.061	-0.016	6-12	0.081	0.032
13	0.957	-18.64	-0.133	-0.055	6-13	0.184	0.010
14	0.926	-19.55	-0.150	-0.050	7-8	-0.0002	-0.210
					7-9	0.269	0.142
					9-10	0.064	-0.010
					9-14	0.086	0.003
					10-11	-0.053	-0.068
					12-13	0.0189	0.0139

which prevents drastic measures taken by the energy system authorities in response to a fault event in a nonessential element in the water system. Table 2 provides the estimated and measured numerical values of the associated parameters in both water and energy systems. Bad data is pointed out with red checkmarks, where detected, and the associated errors are determined with red color.

5.3. False data injection on the pump P_R 's flow

In this case, an arbitrary FDI equal to 74% of the nominal value of q_R is incorporated in the flow q_{PR} , which is deliberately chosen to be large to bring the flow beyond the capacity of the pump P_R , causing multiple failures in the system such as bursting the pipe connected to the pump, damaging the pump, and flooding the downstream node (Shuang et al.,

Table 2
False Data Injection in Pump 2.

Standalone Energy System				
Measurement	Measured	Estimated	Error	Bad Data
	pu	pu	%	
V	1.060	1.11	0.270	X
P_2	0.091	0.142	0.26	X
P_3	-1.56	-1.39	2.75	✓
$P_{1,2}$	1.57	1.60	0.115	X
$P_{2,3}$	0.734	0.911	3.13	✓
$P_{4,2}$	-0.543	0.466	0.576	X

Integrated Water-Energy System				
Measurement	Measured	Estimated	Error%	Bad Data
q_{p1} (LPS)	100	100	0.070	X
q_{p2} (LPS)	63.25	53.55	15.33	✓
q_{p3} (LPS)	26.29	27.15	3.28	X
q_{pR} (LPS)	670	670	0.00	X
q_{pE} (LPS)	119	130	8.2	X
P_{p1} (MW)	6.88	6.884	0.070	X
P_{p2} (MW)	7.48	6.34	15.33	✓
P_{p3} (MW)	2.33	2.41	3.28	X
P_{pR} (MW)	46.07	46.08	0.00	X
P_{pE} (MW)	8.22	8.91	8.5	X
V	1.060	1.11	0.270	X
P_2	0.091	0.142	0.26	X
P_3	-1.56	-1.39	2.75	✓
$P_{1,2}$	1.57	1.60	0.115	X
$P_{2,3}$	0.734	0.911	3.13	✓
$P_{4,2}$	-0.543	0.466	0.576	X

2014). The notion is to compare the performance of the standalone and integrated bad data detection algorithms when a critical component is injected with a large amount of false data. The left subplot in Fig. 7 shows that the standalone energy state estimator detected bad data (shown with red crosses) in the active power injection on bus No.3 (P_3) and the active power flow from bus No.2 to bus No.3 ($P_{2,3}$), which were expected given that bus No.3 supplies power to pump R. Comparing the

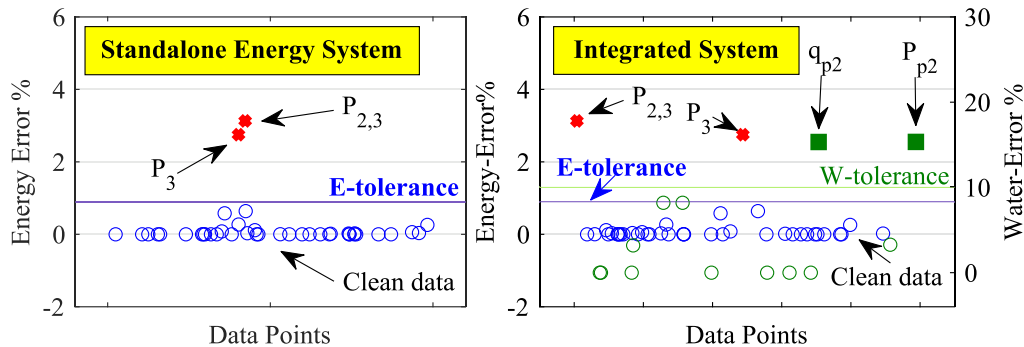


Fig. 6. Bad data detection by the energy state estimator (alone) on the left subplot, and the integrated water-energy state estimator on the right subplot when an FDI of 10 LPS occurred in pump 2 of the water network.

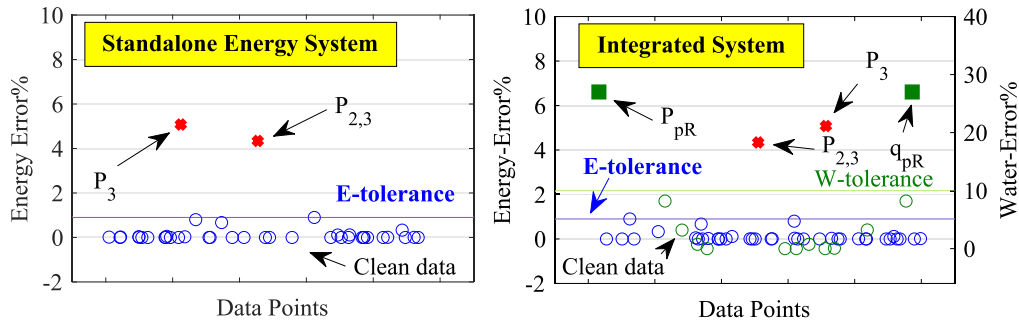


Fig. 7. Bad data detection by the energy state estimator (alone) on the left subplot, and the integrated water-energy state estimator on the right subplot when an FDI of 249 LPS occurred in pump R in the water network.

results with the previous case in Fig. 6, the standalone energy system does not differentiate between these two inherently different FDIs, one pertaining a small injection in a nonessential component whereas the other corresponding to a very large injection in a critical element of the water system. However, on the right subplot, two additional data points shown by green squares are flagged as bad data by the integrated state estimator, identifying pump R as the origin of the FDI and resulting in a large error of 27% in the flow rate and power of the tampered pump which indicates the large amount of the FDI in pump R's measurement. By looking at the left subplots in Figs. 6 and 7, the energy system's operator cannot distinguish between the bad data as the result of FDI in pump 2, which was a minor issue, and of that in pump R, which can be very serious with potentially damaging consequences. On the contrary, in the proposed integrated system (if the right subplots in Figs. 6 and 7 are considered) the differences are indisputable and the origin of each bad data is clearly identified, consequently the energy system's operators will be able to distinguish between an actual critical FDI event requiring drastic measures with a trivial bad data occurrence that could wrongfully be mistaken as critical and does not call for dire decisions. The SCADA on the standalone energy system might bypass bus No.3 resulting in a blackout in loads connected to bus 3, but in the integrated system, only pump R will be removed without affecting other loads. Table 3 presents the estimated and measured values, as well as the estimation errors of the relevant parameters in the system for this case, where bad data are shown with red checkmarks.

5.4. False data injection in the tank's water level

In this case, a drastic fault event in the water system is designed to examine and compare the responses of the standalone energy state estimator and the proposed integrated state-estimation framework. A random false data equal to 18% of the nominal value of water level in the tank (L_T) is injected in the tank level measurements to increase its volume. The goal is to test the efficiency of the standalone and integrated systems in detecting a false data injection in a critical component (i.e., the water tank) of the system that is small enough to not be detected by the SCADA but, if left undetected, it will have damaging effects on the system's operation, e.g., tank overflow. While the level of water in the tank is not directly connected to a pump, it can indirectly affect the flows in pumps P_R and P_E . The mass balance around the tank requires

$$\frac{dV_T}{dt} = q_{PR} - Q_T$$

$$V_T = (\pi/4)(D_T)^2(L_T)$$

Hence

$$\frac{dV_T}{dt} = \left(\pi/4 \right) (D_T)^2 \left(\frac{dL_T}{dt} \right)$$

where V_T is the volume of water in the tank, D_T is the diameter of the tank, L_T is the level of the water in the tank, $\frac{dL_T}{dt}$ is the rate of change in

Table 3

False Data Injection in Pump R.

Standalone Energy System				
Measurement	Measured	Estimated	Error	Bad Data
	pu	pu	%	
V	1.060	1.14	0.676	X
P_2	0.091	0.148	0.329	X
P_3	-1.72	-1.51	4.34	✓
$P_{1,2}$	1.57	1.61	0.145	X
$P_{2,3}$	0.734	0.959	5.08	✓
$P_{4,2}$	-0.543	0.448	0.891	X
Integrated Water-Energy System				
Measurement	Measured	Estimated	Error%	Bad Data
q_{p1} (LPS)	100	100.078	0.078	X
q_{p2} (LPS)	53.25	53.65	0.75	X
q_{p3} (LPS)	26.29	27.13	3.22	X
q_{pR} (LPS)	919	670	27.1	✓
q_{pE} (LPS)	119	129.1	8.2	X
P_{p1} (MW)	6.88	6.88	0.078	X
P_{p2} (MW)	6.304	6.352	0.75	X
P_{p3} (MW)	2.334	2.41	3.23	X
P_{pR} (MW)	63.18	46.07	27.07	✓
P_{pE} (MW)	8.22	8.94	8.6	X
V	1.060	1.14	0.676	X
P_2	0.091	0.148	0.329	X
P_3	-1.72	-1.51	4.34	✓
$P_{1,2}$	1.57	1.61	0.145	X
$P_{2,3}$	0.734	0.959	5.08	✓
$P_{4,2}$	-0.543	0.448	0.891	X

the level of the tank (which is compromised in this case), and Q_T is the water leaving the tank to supply the network. On the other hand,

$$q_E = Q_T - \sum_{k=1}^{12} D_k$$

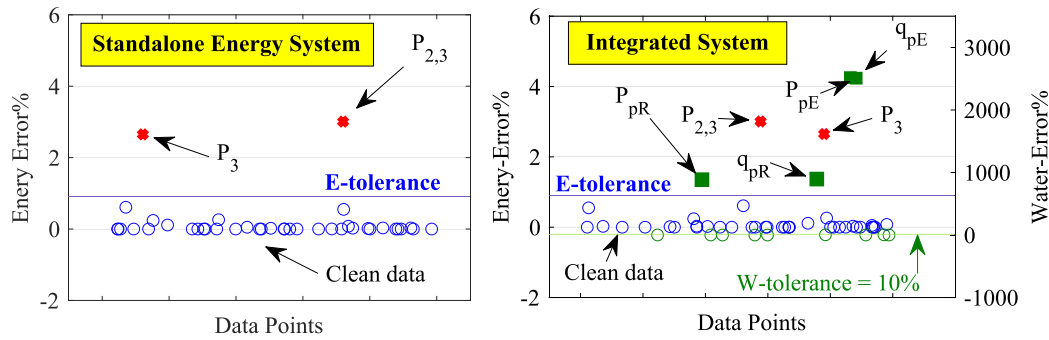


Fig. 8. Bad data detection by the energy state estimator (alone) on the left subplot, and the integrated water-energy state estimator on the right subplot when an FDI of 3 m/s occurred in water level in the tank.

where q_E is the excess flow returned to the reservoir, which is equal to the water injected by the tank to the network minus total usage ($\sum_{k=1}^{12} D_k$). Hence, any change in the water level of the tank can affect both the reservoir and excess flows, passing through P_R and P_E pumps, respectively.

The injected false data is chosen to be large enough to be able to overflow the tank, causing potential flooding in the downstream region and serious damages to the water system infrastructure. The left subplot in Fig. 8 presents the detected bad data by the standalone energy state estimator in P_3 and $P_{2,3}$ as the result of compromised data in pumps R and E, both of which, receive power from bus No.3. However, no further information to the extent of the danger and the origin of such manipulation is provided. In fact, there is no difference between the data observed in any of the left subplots in Figs. 6–8, since the WDS is contemplated as a load like other loads connected on bus No.3. Hence, the operational decision towards any of these FDIs, regardless of the origin and degree of importance, would be the same (e.g., disconnecting all loads on bus No.3). However, the results of the proposed integrated state estimator in the right subplots can provide valuable remarks to an operator on identifying the root cause of the event and resolving it as soon as possible without having to shut down the entire supply on bus No.3. While the left subplot exhibits results similar to those reported in the previous case studies, the right subplot in Fig. 8 demonstrates very large errors in the flow and power of two essential pumps in the water system (more than 1000%), as well as in the generator providing the power, indicating an issue beyond one pump's malfunction. This outcome discloses the irrefutable impact of the integrated bad data detection algorithm in managing a combined water-energy system. Not only will it prevent unnecessary downtime and power outage in the combined system, which will also offset the cost of restarting operations in both water and energy systems, but also will accurately ascertain critical events when a key element is manipulated or is on the verge of failure. The values presented in Table 4 show the large errors detected in these parameters. The table also provides all the estimated and measured data of the relevant components of the studied system, pointing out bad data and high errors were detected.

5.5. Remarks

This subsection summarizes some of the advantages of the proposed integrated state-estimation framework for interdependent water and energy systems. The main idea behind an integrated state-estimation framework is to enhance the security of the system. Currently, water and energy networks are treated separately. For example, assume an energy network that is run independently. The loads in an energy network are normally considered as constant power loads and smart meters are dedicated to monitor loads in different buses (Nodes). An attacker can directly inject false data into load measurements without being detected by state-estimation. However, for a successful attack, the attacker needs to manipulate measurements in vicinity of targeted node

Table 4

False Data Injection in the Tank Level.

Standalone Energy System				
Measurement	Measured pu	Estimated pu	Error %	Bad Data
V	1.060	1.11	0.25	X
P_2	0.091	0.142	0.258	X
P_3	-1.55	-1.38	2.65	✓
$P_{1,2}$	1.571	1.60	0.112	X
$P_{2,3}$	0.734	0.907	3.01	✓
$P_{4,2}$	-0.543	0.468	0.555	X
Integrated Water-Energy System				
Measurement	Measured	Estimated	Error%	Bad Data
q_{p1} (LPS)	100.0	100.08	0.078	X
q_{p2} (LPS)	53.25	53.65	0.75	X
q_{p3} (LPS)	26.29	27.13	3.22	X
q_{pR} (LPS)	670	6670	895	✓
q_{pE} (LPS)	119	3132	2518	✓
P_{p1} (MW)	6.88	6.88	0.078	X
P_{p2} (MW)	6.304	6.352	0.75	X
P_{p3} (MW)	2.334	2.41	3.23	X
P_{pR} (MW)	46.08	54.03	17.25	✓
P_{pE} (MW)	8.22	214.7	2509	✓
V	1.060	1.11	0.25	X
P_2	0.091	0.142	0.258	X
P_3	-1.55	-1.38	2.65	✓
$P_{1,2}$	1.571	1.60	0.112	X
$P_{2,3}$	0.734	0.907	3.01	✓
$P_{4,2}$	-0.543	0.468	0.555	X

as well (Yuan et al., 2011). In an integrated framework, an attacker needs to manipulate more measurements in order to have a successful attack. Fig. 9 has been added as an example to visualize this concept. Consider the left-side plot, an attacker tries to cause harm on an independent energy network by injecting false data on a load terminal. To

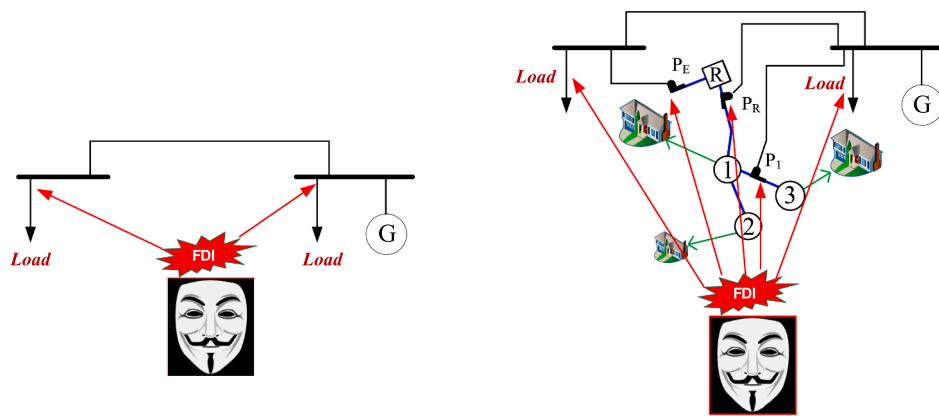


Fig. 9. Concept of false data injections in standalone energy network and integrated water-energy network.

launch a successful attack, the attacker may need to inject false data on both load terminals as they are physically connected. Now, assume an integrated water-energy network in the right-side plot. In order to have a successful attack, the attacker needs to inject false data on five measurements. Compared to a standalone system, the attacker will have a much harder time to launch a successful attack. Therefore, the integrated framework inherently helps security of the system against false data injection (FDI) attacks.

In addition, it was demonstrated through case studies that an integrated state-estimation framework can improve the state-estimation and bad data detection process in water and energy networks. Using available measurements in water distribution networks and existing Phasor Measurement Units (PMUs) in electricity distribution networks, the proposed framework can easily be incorporated in Supervisory Control and Data Acquisition (SCADA) centers to run in real-time, enhance the resilience of water-energy networks, and detect anomalies.

6. Conclusion

An integrated framework was proposed in this paper to estimate the parameters of a combined water-energy network and detect potential false data injections. The main novelty of the proposed integrated state-estimation framework is that a central control system is able to not only detect bad data across the integrated water-energy system, but also identify the exact location of the fault event occurrence. The findings of the paper indicate that in the absence of such integrated detection framework, the power of the entire water distribution, as well as other residential loads, depending on the scale of the compromised bus, could be disconnected due to bad data detection in the power network. However, the proposed framework can identify the location of the tampered data and the component responsible for the false data injection. This hypothesis is tested with several case studies and the results demonstrate that a standalone energy system fails to distinguish between the false data injections on the tank's water level and flow of pumps in the system, and thus the operator may decide to disconnect the power to the load entirely (water distribution in this case) resulting

in a blackout in a critical infrastructure. On the other hand, the proposed integrated system detects false data injections on pumps differently from those on the tanks' water levels such that the operator could avoid blackouts by isolating the attacked equipment/readings, therefore, the resilience of the integrated water-energy network will significantly increase using the proposed integrated framework. The outcome of this work will help developing the future research focusing on various types of false data injections including cyberattacks and detection of threats in an integrated water-energy network. One of the limitations of the proposed integrated model is that it may not be computationally efficient for large systems, as it would involve numerous iterations. In future research, the models need to be scaled up and adjusted to detect stealthy false data injections in large scale water-energy system using big data analytic methods.

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CRediT authorship contribution statement

Faegheh Moazeni: Conceptualization, Methodology, Software, Writing - review & editing, Investigation, Visualization. **Javad Khazaei:** Conceptualization, Methodology, Software, Writing - review & editing, Investigation, Visualization. **Prasenjit Mitra:** Investigation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

The variables in a water-energy network include measurements and constant parameters of the system that are adopted from the existing benchmarks, and state-variables that should be solved by the state-estimation process. For instance, nodal heads and demands are considered to be the state variables in water systems, while the voltage magnitude and angle, active and reactive power injections, and active and reactive power flows are the state variables of the energy systems to be solved as the output of the state-estimation models. The set of data used as the given measurements in the case studies are adopted from standard database of the referenced water and energy benchmarks as listed in Table A1 (Hernandez et al., 2016). Parameters of the energy network were adopted from MATPOWER toolbox of MATLAB (Zimmerman et al., 2010).

Table A1
Parameters of the Water System (Bargiela, 1984; Hernadez et al., 2016).

Parameter	Values	Parameter	Values
Q_T	335LPS	q_E	1.19LPS
H_T	59 m	q_{p1}	100 m
q_R	335LPS	q_{p2}	53.25LPS
H_R	20 m	q_{p3}	26.29LPS
Pumps' characteristic equations			
Pump No.		A	B
1		86.00	$0.06286q_p^{1.41}$
2		111.00	$0.01914q_p^{2.05}$
3		148.00	$0.0001419q_p^{2.05}$
R		86.00	$0.06286q_p^{1.41}$
E		86.00	$0.06286q_p^{1.41}$
Pipe from/to Node	Diameter m	Length m	Roughness
T/1	0.7	610	120
1/2	0.305	610	110
3/2	0.254	610	115
4/3	0.203	1219	100
5/4	0.203	1219	110
6/5	0.254	610	100
7/6	0.381	607	110
3/9	0.381	1097	135
11/10	0.457	1930	110
2/11	0.305	3151	100
1/12	0.457	762	110
4/8	0.229	914	125
10/3	0.305	823	140
9/5	0.154	411	100
8/6	0.229	701	110
12/11	0.152	1072	135
2/4	0.152	864	90
8/1	0.152	711	90
5/7	0.152	832	100

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